

1. Administrative & Study Identification

Full Study Title: A Phase 2 Study to Evaluate the Efficacy and Safety of MK-1026 (Nemtabrutinib) in Participants With Hematologic Malignancies (MK-1026-003) **Short Title/Acronym:** BELLWAVE-003 **Protocol Number:** MK-1026-003 **NCT Number:** NCT04728893 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Nemtabrutinib (MK-1026, formerly ARQ 531) **Phase of Development:** Phase II **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and efficacy of nemtabrutinib (MK-1026) in participants with relapsed or refractory (R/R) hematologic malignancies and to determine the Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To assess the Objective Response Rate (ORR), Duration of Response (DOR), Progression-Free Survival (PFS), and Overall Survival (OS) across disease-specific cohorts.

2.2 Background & Rationale Treatment options are limited for patients with certain hematologic malignancies (such as CLL/SLL) who have relapsed or refractory disease after failing standard therapies, including Bruton tyrosine kinase (BTK) inhibitors and BCL2 inhibitors. Acquired resistance is frequently driven by BTK C481S mutations. Nemtabrutinib is a noncovalent, reversible BTK inhibitor with a unique kinase profile designed to target both wild-type and C481S-mutant BTK, addressing a critical unmet medical need in this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (within multiple disease-specific expansion cohorts) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 450 participants (Part 1: ~30 participants; Part 2: ~420 participants) **Number of Sites:** Global and Regional multicenter study **Estimated Study Start:** 05-Apr-2021 **Estimated Primary Completion:** 04-Jan-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 within 7 days prior to allocation. 2. Confirmed diagnosis of relapsed or refractory (R/R) hematologic malignancies including CLL/SLL, Richter's transformation, MZL, MCL, FL, or Waldenström's macroglobulinemia (WM). 3. Life expectancy of at least 3 months, and the ability to swallow and retain oral medication.

4.2 Exclusion Criteria 1. Active central nervous system (CNS) disease. 2. Active HBV/HCV infection (unless well-controlled under specific antiviral thresholds) or an active infection requiring systemic therapy. 3. History of other malignancies $\leq$ 3 years before providing informed consent (exceptions apply for curatively treated localized cancers, e.g., basal cell carcinoma).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Nemb Brutinib tablets administered orally once daily (QD). The study explores dose escalation (Part 1) to determine the RP2D (e.g., 65 mg QD), followed by dosing at the RP2D across 8 disease-specific expansion cohorts (Part 2). **Route of Administration:** Oral **Duration of Treatment:** Until disease progression or unacceptable toxicity (up to 71 months for the expansion cohort monitoring).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for oncology patients. **Prohibited:** Receipt of prior systemic anti-cancer therapy or investigational agents within 4 weeks prior to the first dose of study intervention.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Biomarker/Imaging assessments
Baseline	Day 0	Cohort Allocation, First Dose, Safety Labs
Interim	Regular Cycles	Efficacy Assessments, Dose-limiting toxicity (DLT) evaluation, Safety Labs
End of Study	Variable	Final Tumor Assessment, Exit Interview, IP Accountability, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Objective Response Rate (ORR) and determination of the Recommended Phase 2 Dose (RP2D) based on Dose-Limiting Toxicities (DLTs). **Measurement Method:** Evaluated via Investigator and Independent Central Review using disease-

specific response criteria (e.g., Lugano Classification or iwCLL criteria).

7.2 Secondary & Exploratory Endpoints * Duration of Response (DOR) * Progression-Free Survival (PFS) * Overall Survival (OS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous AE collection evaluating toxicity grade according to NCI CTCAE. **Serious Adverse Events (SAEs):** Safety profile continuously monitored; reporting timeline strictly typically 24 hours. **Data Monitoring Committee (DMC):** Internal and external safety review utilized particularly during the Part 1 dose escalation and confirmation phases.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety and efficacy will be assessed in all participants who receive ≥ 1 dose of the study drug (Safety Set). **Sample Size Justification:** ~450 participants. ORR and 95% Confidence Intervals (CI) will be estimated using an exact binomial method to ensure statistical robustness. **Handling of Missing Data:** Time-to-event endpoints (DOR, PFS, OS) will be analyzed and estimated using the Kaplan-Meier method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring per Sponsor auditing protocols. **Audit Trail:** Electronic Case Report Form (eCRF) tracking with Central Imaging reviews to assure independent bias-free assessment.

11. Study Identifiers & Governance

Primary ID: MK-1026-003 **Registry Number:** NCT04728893 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** BELLWAVE-003 **Official Regulatory Title:** A Phase 2 Study to Evaluate the Efficacy and Safety of MK-1026 in Participants With Hematologic Malignancies

12. Clinical Framework (Oncology Specific)

Primary Condition: Hematologic Malignancies (CLL/SLL, MZL, MCL, FL, WM) **Diagnosis Threshold:** Confirmed Relapsed/Refractory Disease (e.g., progression post-BTK/BCL2 inhibitor therapy)

Medical Methodology: Targeted Therapy (Noncovalent BTK inhibitor)
Optimization Target: Objective response, tumor shrinkage, and halting disease progression

13. Study Procedures & Methodology

Management Pathway: Targeted oral administration with dose escalation followed by broad cohort expansion
Education Protocol (HCP): Site Initiation Visits (SIV) and strict protocol training for DLT reporting
Education Protocol (Patient): Informed consent review, pill-swallowing compliance, and self-reporting AE instructions
Quality Audit Mechanism: Independent central efficacy review and central laboratory biomarker testing

14. Observation & Follow-Up Schedule

Total Duration: Up to ~71 months for evaluation of long-term PFS and OS
Onsite Visit Cadence: Determined by 28-day treatment cycles
Remote Monitoring Frequency: Continuous survival follow-up per protocol
Checklist Review Interval: At the start of every new treatment cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				